

Phase 1.5: Maine Yield-Curve Stratification (v2)

Companion to LANDOWNER_INTEGRATION_STRATEGY.md addendum on yield curves as the cross-model integration backbone. Builds the 6 forest-type × 3 ecoregion × 6 owner-class = 108-cell lattice, with FIA plot counts per cell. Yield curves fit per populated cell will become the primary input for GCBM, LANDIS-II, and CEM, with FVS as the source-of-truth generator.

Method

Stratification axes:

- Forest type (6 buckets): Spruce-fir (FORTYPECD 121–128), Northern hardwood (800–809), Aspen-birch (900–919), Mixedwood (700–799), White/Red pine (100–119), Oak/Pine/Hemlock (400–599), Other.
- Ecoregion (3 zones, libcbm AIDB-aligned): ME_NH (Northern Highlands), ME_NCZ (Northern Central Zone), ME_APH (Acadian Plains/Hills).
- Owner (6 HCB classes): NIPF, Industrial, Tribal, Federal, State, Local. Non-Forest and Water raster classes are masked out.

Result

Of 108 theoretical cells, 105 are populated and **32 cells have $n \geq 30$ plots**. The well-sampled cells cover 4,434 of 5,027 Maine forest plots (88 percent of the population). Sparse cells will need to be collapsed before yield-curve fitting; the natural collapse is Tribal + Federal + Local → Public-Other.

Top 32 cells ($n \geq 30$)

Forest type	Ecoregion	Owner	n plots	Status
Northern hardwood	ME_APH	NIPF	338	ok
Northern hardwood	ME_NCZ	NIPF	334	ok
Spruce-fir	ME_NH	Industrial	329	ok
Spruce-fir	ME_APH	NIPF	326	ok
Spruce-fir	ME_NH	NIPF	300	ok
Northern hardwood	ME_NH	Industrial	281	ok
Other	ME_APH	NIPF	275	ok
Northern hardwood	ME_NH	NIPF	243	ok
Northern hardwood	ME_NCZ	Industrial	176	ok
Spruce-fir	ME_NCZ	NIPF	163	ok
White/Red pine	ME_APH	NIPF	152	ok

Forest type	Ecoregion	Owner	n plots	Status
Spruce-fir	ME_APH	Industrial	143	ok
Northern hardwood	ME_APH	Industrial	123	ok
Spruce-fir	ME_NCZ	Industrial	123	ok
Aspen-birch	ME_APH	NIPF	107	ok
Oak/Pine/Hemlock	ME_APH	NIPF	92	ok
Other	ME_APH	Industrial	92	ok
Aspen-birch	ME_NH	NIPF	77	ok
Aspen-birch	ME_NH	Industrial	76	ok
Other	ME_NCZ	NIPF	70	ok
Spruce-fir	ME_NH	State	70	ok
Aspen-birch	ME_NCZ	NIPF	67	ok
Northern hardwood	ME_NCZ	State	64	ok
Other	ME_NH	NIPF	61	ok
Aspen-birch	ME_APH	Industrial	53	ok
Northern hardwood	ME_NH	State	52	ok
White/Red pine	ME_APH	Industrial	50	ok
Aspen-birch	ME_NCZ	Industrial	49	ok
Mixedwood	ME_APH	NIPF	42	ok
White/Red pine	ME_NCZ	NIPF	38	ok
Other	ME_NH	Industrial	37	ok
Spruce-fir	ME_NCZ	State	31	ok

Cells in bold (n ≥ 100) are the priority for first-pass yield-curve fitting. Northern hardwood and Spruce-fir × NIPF/Industrial dominate the well-sampled portion; spatial coverage spans all three ecoregions.

Next steps

1. Collapse Tribal/Federal/Local → Public-Other to lift sparse cells over n=30.
2. Run FVS-NE and FVS-Acadian factorial across the populated cells × 4 treatment trajectories (notreat, light_partial, heavy_partial, clearcut_regen) on Cardinal.
3. Fit Chapman-Richards ($y = a \times (1 - \exp(-b \times \text{age}))^c$) for AGB and flexible piecewise for treated trajectories where rotation imposes discontinuities.

4. Archive curves as `maine_yield_curves_v1.csv` (long format) plus `.parquet` for fast ingestion. Include 5/50/95 quantile columns.

5. Write thin model-specific adapters: GCBM ingestion, LANDIS PnET parameter writer, CEM stratification CSV, and Woodstock periodic-volume CSV.

Files

- `config/maine_yield_curve_strata.csv` — the 108-cell lattice with FIA plot counts
- `scripts/build_yield_curve_strata.R` — generator script
- `docs/LANDOWNER_INTEGRATION_STRATEGY.md` — design rationale (uploaded 28 April)
- `docs/FVS_WOODSTOCK_STRATEGY.md` — full strategy for cross-model yield-curve generation + Woodstock optimization (this companion doc, separate file)